# Chapter 2 - The Real Number System

Note 2

Lecturer: Dr. Rajitha Ranasinghe Academic Year: 2024/2025

## 2.2 : Ordered Field Axioms

In this section, we explore the algebraic structure of the real number system.We shall assume that the set of real numbers $\mathbb{R}$ satisfies the following axioms, which define its algebraic structure as a **field**.

**Axiom Set 1 - (FIELD AXIOMS)**

There are functions $+$ and $\cdot$, defined on $\mathbb{R}^2 := \mathbb{R} \times \mathbb{R}$, which satisfy the following properties for every $a, b, c \in \mathbb{R}$ :

- **Closure Properties :** $a + b$ and $a \cdot b$ belongs to $\mathbb{R}$.
- **Associative Properties :** $a + (b + c) = (a + b) + c$ and $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.
- **Commutative Properties :** $a + b = b + a$ and $a \cdot b = b \cdot a$.
- **Distributive Law :** $a \cdot (b + c) = a \cdot b + a \cdot c$.
- **Existence of the Additive identity :** There is a unique element $0 \in \mathbb{R}$ such that $0 + a = a$ for all $a \in \mathbb{R}$.
- **Existence of the Multiplicative identity :** There is a unique element $1 \in \mathbb{R}$ such that $1 \neq 0$ and $1 \cdot a = a$ for all $a \in \mathbb{R}$.
- **Existence of the Additive inverses :** For every $x \in \mathbb{R}$ there is a unique element $-x \in \mathbb{R}$ such that
$$x + (-x) = 0.$$
- **Existence of the Multiplicative inverses :** For every $x \in \mathbb{R} \setminus \{0\}$ there is a unique element $x^{-1} \in \mathbb{R}$ such that
$$x \cdot (x^{-1}) = 1.$$

**Note :**

- We denote $a + (-b)$ by $a - b$; $a \cdot b$ by $ab$; $a^{-1}$ by $\frac{1}{a}$ or $1/a$, and $a \cdot b^{-1}$ by $\frac{a}{b}$ or $a/b$. Notice that by the existence of additive and multiplicative inverses, the equation $x + a = 0$ can be solved for each $a \in \mathbb{R}$, and the equation $ax = 1$ can be solved for each $a \in \mathbb{R}$ provided that $a \neq 0$.

- From the **Axiom Set 1**, we can derive all the usual algebraic laws of real numbers, including the following:

$$(-1)^2 = 1, \tag{1}$$

$$0 \cdot a = 0, \quad -a = (-1) \cdot a, \quad -(-a) = a, \quad a \in \mathbb{R}, \tag{2}$$

$$-(a - b) = b - a, \quad a, b \in \mathbb{R}, \tag{3}$$

and

$$a, b \in \mathbb{R} \text{ and } ab = 0 \text{ imply } a = 0 \text{ or } b = 0. \tag{4}$$

The Field Axioms are sufficient to derive all **algebraic properties** of $\mathbb{R}$, but it does not completely describe the real number system. The set of real numbers also has an order relation (i.e., a concept of "less than").

## Axiom Set 2 - ORDER AXIOMS

There is a relation '$<$' on $\mathbb{R} \times \mathbb{R}$ that has the following properties:

- **Trichotomy Property** Given $a, b \in \mathbb{R}$, one and only one of the following statements holds:

$$a < b, \quad b < a, \quad \text{or } a = b.$$

- **Transitive Property.** For $a, b, c \in \mathbb{R}$,

$$a < b \text{ and } b < c \text{ imply } a < c.$$

- **The Additive Property.** For $a, c, c \in \mathbb{R}$,

$$a < b \text{ and } c \in \mathbb{R} \text{ imply } a + c < b + c.$$

- **The Multiplicative Properties.** For $a, b, c \in \mathbb{R}$,

$$a < b \text{ and } c > 0 \text{ imply } ac < bc$$

and

$$a < b \text{ and } c < 0 \text{ imply } bc < ac.$$

By $b > a$ we shall mean $a < b$. By $a \leq b$ and $b \geq a$ we shall mean $a < b$ or $a = b$. By $a < b < c$ we shall mean $a < b$ and $b < c$. In particular, $2 < x < 1$ makes no sense at all.

**Note :** There are two *Multiplicative Properties*, so every time you multiply an inequality by an expression, you must carefully note the sign of that expression and adjust the inequality accordingly. For example, $x < 1$ does NOT imply that $x^2 < x$ unless $x > 0$. If $x < 0$, then by the Second Multiplicative Property, $x < 1$ implies $x^2 > x$.

The real number system $\mathbb{R}$ contains certain special subsets: the set of **Natural numbers**

$$\mathbb{N} := \{0, 1, 2, 3, \dots\},$$

obtained by beginning with 1 and successively adding 1s to form $2 := 1 + 1, 3 := 2 + 1$, and so on; the set of *integers*

$$\mathbb{Z} := \{..., -2, -1, 0, 1, 2, ...\}$$

(*Zahl* is German for number); the set of *rationals* (or fractions or **Q**uotients)

$$\mathbb{Q} := \left\{ \frac{m}{n} : m, n \in \mathbb{Z} \text{ and } n \neq 0 \right\};$$

and the set of *irrationals*

$$\mathbb{Q}^c = \mathbb{R} \setminus \mathbb{Q}.$$

Equality in $\mathbb{Q}$ is defined by

$$\frac{m}{n} = \frac{p}{q} \quad \text{if and only if} \quad mq = np.$$

Recall that each of the sets $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$, and $\mathbb{R}$ is a proper subset of the next; that is

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}.$$

For example, every rational number is a real number (because $\frac{m}{n} := mn^{-1}$ is a real number by postulate 1), but $\sqrt{2}$ is an irrational.

**Remark :** We will assume that the sets $\mathbb{N}$ and $\mathbb{Z}$ satisfy the following properties:

(i). If $n, m \in \mathbb{Z}$, then $n + m$, $n - m$, and $mn$ belongs to $\mathbb{Z}$.

(ii). If $n \in \mathbb{Z}$, then $n \in \mathbb{N}$ if and only if $n \geq 0$.

(iii). There is no $n \in \mathbb{Z}$ that satisfies $0 < n < 1$.

**Note :** Using these properties, we can prove that $\mathbb{Q}$ satisfies the Field Axioms.

**Example 1 :** If $a \in \mathbb{R}$, prove that

$$a \neq 0 \text{ implies } a^2 > 0.$$

In particular, $-1 < 0 < 1$.

***Proof.***

- Suppose that $a \neq 0$. By the Trichotomy Property, either $a > 0$ or $a < 0$.
- *Case 1.* $\boxed{a > 0}$ Multiply both sides of this inequality by $a$, using the First Multiplicative Property. We obtain $a^2 = a \cdot a > 0 \cdot a$. Since (by (2)), $0 \cdot a = 0$ we conclude that $a^2 > 0$.
- *Case 2.* $\boxed{a < 0}$ Multiply both sides of this inequality by $a$. Since $a < 0$, it follows from the Second Multiplicative Property that $a^2 = a.a > 0.a = 0$.
- Since $1 \neq 0$, it follows that $1 = 1^2 > 0$. Adding $-1$ to both sides of this inequality, we conclude that $0 = 1 - 1 > 0 - 1 = -1$.

□

**Example 2 :** If $a \in \mathbb{R}$, prove that

$$0 < a < 1 \text{ implies } 0 < a^2 < a \text{ and } a > 1 \text{ implies } a^2 > a.$$

***Proof.***

- Suppose that $0 < a < 1$. Multiply both sides of the inequality by $a$ using the First Multiplicative Property. We obtain $0 = 0 \cdot a < a^2 < 1 \cdot a = a$. In particular, $0 < a^2 < a$.
- On the other hand, if $a > 1$, then $a > 0$ by **Example 1** and the Transitive Property. Multiplying $a > 1$ by $a$, we conclude that $a^2 = a \cdot a > 1 \cdot a = a$.

□

Similarly, we can prove that

$$0 \leq a < b \text{ and } 0 \leq c < d \text{ imply } ac < bd, \tag{5}$$

$$0 \leq a < b \text{ implies } 0 \leq a^2 < b^2 \text{ and } 0 \leq \sqrt{a} < \sqrt{b}, \tag{6}$$

and

$$0 < a < b \text{ implies } \frac{1}{a} > \frac{1}{b} > 0. \tag{7}$$

Much of the analysis deals with estimation (of error, of growth, of volume, etc.) in which these inequalities and the following concept play a central role.

**Definition :** The absolute value of a number $a \in \mathbb{R}$ is the number

$$|a| := \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$$

When providing results about the absolute value, we can always break the proof up into several cases, depending on when the parameter is positive, negative, or zero.

**Remark :** The absolute value is multiplicative; that is, $|ab| = |a||b|$ for all $a, b \in \mathbb{R}$.

The following result is useful when solving inequalities involving absolute value signs.

**Theorem 2.1 : (Fundamental Theorem of Absolute Values)**
Let $a \in \mathbb{R}$ and $M \geq 0$. Then $|a| \leq M$ if and only if $-M \leq a \leq M$.

***Proof.***

- Suppose first that $|a| \leq M$. Multiplying by $-1$, we also have $-|a| \geq -M$.

  *Case 1*: $\boxed{a \geq 0}$ By the definition, $|a| = a$. Thus by the hypothesis,

  $$-M \leq 0 \leq a = |a| \leq M.$$

  *Case 2.* $\boxed{a < 0}$ By the definition, $|a| = -a$. Thus by the hypothesis,

  $$-M \leq -|a| = a < 0 \leq M.$$

  This proves that $-M \leq a \leq M$ in either case.

- Conversely, if $-M \leq a \leq M$, then $a \leq M$ and $-M \leq a$. Multiplying the second inequality by $-1$, we have $-a \leq M$. Consequently, $|a| = a \leq M$ if $a \geq 0$, and $|a| = -a \leq M$ if $a < 0$.

□

**Note :** In a similar way we can prove that $|a| < M$ if and only if $-M < a < M$.

Here is another useful result about absolute values.

**Theorem 2.2 :** The absolute value satisfies the following three properties.

(i). (**Positive Definite**) For all $a \in \mathbb{R}$, $|a| \geq 0$ with $|a| = 0$ if and only if $a = 0$.

(ii). (**Symmetric**) For all $a, b \in \mathbb{R}$, $|a - b| = |b - a|$.

(iii). (**Triangle Inequalities**) For all $a, b \in \mathbb{R}$,

$$|a + b| \leq |a| + |b| \quad \text{and} \quad ||a| - |b|| \leq |a - b|.$$

***Proof.***

(i). If $\boxed{a \geq 0}$, then $|a| = a \geq 0$. If $\boxed{a < 0}$, then by the definition and the Second Multiplicative Property, $|a| = -a = (-1)a > 0$. Thus $|a| \geq 0$ for all $a \in \mathbb{R}$.

If $|a| = 0$, then by definition $a = |a| = 0$ when $a \geq 0$ and $a = -|a| = 0$ when $a < 0$. Thus $|a| = 0$ implies that $|a| = 0$. Conversely, $|0| = 0$ by the definition.

(ii). From the properties of absolute values, it follows that

$$|a - b| = |-1||b - a| = |b - a|$$

(iii). To prove the first inequality, notice that $|x| \leq |x|$ holds for any $x \in \mathbb{R}$. Thus, **Theorem 2.1** implies that $-|a| \leq a \leq |a|$ and $-|b| \leq b \leq |b|$ . Adding these inequalities, we obtain

$$-(|a| + |b|) \leq a + b \leq |a| + |b|.$$

Hence, again by **Theorem 2.1**, it follows that $|a + b| \leq |a| + |b|$.

To prove the second inequality, apply the first inequality to $(a - b) + b$. We obtain

$$|a| - |b| = |a - b + b| - |b| \leq |a - b| + |b| - |b| = |a - b|.$$

By reversing the roles of $a$ and $b$ and applying part (ii), we also obtain

$$|b| - |a| \leq |b - a| = |a - b|.$$

Multiplying this last inequality by $-1$ and combining it with the preceding one verifies

$$-|a - b| \leq |a| - |b| \leq |a - b|.$$

We conclude by **Theorem 2.1** that $||a| - |b|| \leq |a - b|$.

□

**Note :** The last inequality implies that $|a| - |b| \leq |a - b|$ for all $a, b \in \mathbb{R}$.

**WARNING :** Some students mistakenly mix absolute values and the Additive Property to conclude that $b < c$ implies $|a+b| < |a+c|$. This implication is false unless both $a+b$ and $a+c$ are non-negative. For example, if $a = 1$, $b = -5$, and $c = -1$, then $b < c$ but $|a + b| = 4$ is not less than $|a + c| = 0$.

A correct way to estimate using absolute value signs usually involves one of the triangle inequalities.

**Example 3 :** Prove that if $-2 < x < 1$, then $|x^2 - x| < 6$.

***Proof.*** By hypothesis, $|x| < 2$. Hence by the triangle inequality, it follows that

$$|x^2 - x| \leq |x|^2 + |x| < 4 + 2 = 6.$$

□

The following result (which is equivalent to the **Trichotomy Property**) will be used many times in this and subsequent chapters.

**Theorem 2.3 :** Let $x, y, a \in \mathbb{R}$.

(i). $x < y + \epsilon$ for all $\epsilon > 0$ if and only if $x \leq y$.

(ii). $x > y - \epsilon$ for all $\epsilon > 0$ if and only if $x \geq y$.

(iii). $|a| < \epsilon$ for all $\epsilon > 0$ if and only if $a = 0$.

***Proof.***

(i). Suppose to the contrary that $x < y + \epsilon$ for all $\epsilon > 0$ but $x > y$. Set $\boxed{\epsilon_0 = x - y > 0}$ and observe that $y + \epsilon_0 = x$ . Hence by the Trichotomy Property, $y + \epsilon_0 \leq x$. This **contradicts** the hypothesis for $\epsilon = \epsilon_0$. Thus $x \leq y$.

Conversely, suppose that $x \leq y$. and $\epsilon > 0$ is given. Either $x < y$ or $x = y$. If $x < y$, then $x + 0 < y + 0 < y + \epsilon$ by the Additive and Transitive Properties. If $x = y$, then $x < y + \epsilon$ by the Additive Property. Thus $x < y + \epsilon$ for all $\epsilon > 0$ in either case. This completes the proof of part (i).

(ii). Suppose that $x > y - \epsilon$ for all $\epsilon > 0$. By the Second Multiplicative Property, this is equivalent to $-x < -y + \epsilon$ , hence by part (i), equivalent to $-x \leq -y$. By the Second Multiplicative Property, this is equivalent to $x \geq y$.

(iii). Suppose that $|a| < \epsilon = 0 + \epsilon$ for all $\epsilon > 0$. By part (i), this is equivalent to $|a| \leq 0$ . Since it is always the case that $|a| \geq 0$, we conclude by the Trichotomy Property that $|a| = 0$. Therefore, $a = 0$ by Theorem 2.2 (i).

□

Let $a$ and $b$ be real numbers. A **closed interval** is a set of the form

$$[a,b] := \{x \in \mathbb{R} : a \leq x \leq b\}, \quad [a,\infty) := \{x \in \mathbb{R} : a \leq x\},$$

$$(-\infty, b] := \{x \in \mathbb{R} : x \leq b\}, \quad \text{or} \quad (-\infty,\infty) := \mathbb{R}.$$

and an **open interval** is a set of the form

$$(a,b) := \{x \in \mathbb{R} : a < x < b\}, \quad (a,\infty) := \{x \in \mathbb{R} : a < x\},$$

$$(-\infty, b) := \{x \in \mathbb{R} : x < b\}, \quad \text{or} \quad (-\infty,\infty) := \mathbb{R}.$$

By an **interval** we mean a closed interval, an open interval, or a set of the form

$$[a,b) := \{x \in \mathbb{R} : a \leq x < b\} \quad \text{or} \quad (a,b] := \{x \in \mathbb{R} : a < x \leq b\}.$$

**Note :** When $a < b$, the intervals $[a,b], [a,b), (a,b]$, and $(a,b)$ corresponding to line segments on the real line, but when $b < a$, these "intervals" are all the empty sets.

- An interval $I$ is said to be **bounded** if and only if it has the form $[a,b], (a,b), [a,b)$, or $(a,b]$ for some $-\infty < a \leq b < \infty$, in which case the numbers $a, b$ will be called the **endpoints** of $I$. All other intervals will be called **unbounded**.
- An interval with endpoints $a, b$ is called **degenerate** if $a = b$ and **nondegenerate** if $a < b$. Thus a degenerate open interval is the empty set, and a degenerate closed interval is a point.

.....

## EXERCISES

1. Let $a, b, c, d \in \mathbb{R}$ and consider each of the following statements. Decide which are true and which are false. Prove the true ones and give counterexamples to the false ones.

   a) If $a < b$ and $c < d < 0$, then $ac > bd$.

   b) If $a \leq b$ and $c > 1$, then $|a + c| \leq |b + c|$.

   c) If $a \leq b$ and $b \leq a + c$, then $|a - b| \leq c$.

   d) If $a < b - \varepsilon$ for all $\varepsilon > 0$, then $a < 0$.

2. Suppose that $a, b, c \in \mathbb{R}$ and $a \leq b$.

   a) Prove that $a + c \leq b + c$.

   b) If $c \geq 0$, prove that $a \cdot c \leq b \cdot c$.

3. Prove (5), (6), and (7). Show that each of these statements is false if the hypothesis $a \geq 0$ or $a > 0$ is removed.

4. The positive part of an $a \in \mathbb{R}$ is defined by

$$a^+ := \frac{|a| + a}{2}$$

   and the negative part by

$$a^- := \frac{|a| - a}{2}$$

   a) Prove that $a = a^+ - a^-$and $|a| = a^+ + a^-$.

   b) Prove that

$$a^+ = \begin{cases} a & a \geq 0 \\ 0 & a \leq 0 \end{cases} \quad \text{and} \quad a^- = \begin{cases} 0 & a \geq 0 \\ -a & a \leq 0. \end{cases}$$

5. Solve each of the following inequalities for $x \in \mathbb{R}$.

   (a) $|2x + 1| < 7$

   (b) $|2 - x| < 2$

   (c) $|x^3 - 3x + 1| < x^3$

   (d) $\dfrac{x}{x - 1} < 1$

   (e) $\dfrac{x^2}{4x^2 - 1} < \dfrac{1}{4}$

6. Let $a, b \in \mathbb{R}$.

   a) Prove that if $a > 2$ and $b = 1 + \sqrt{a-1}$, then $2 < b < a$.

   b) Prove that if $2 < a < 3$ and $b = 2 + \sqrt{a-2}$, then $0 < a < b$.

   c) Prove that if $0 < a < 1$ and $b = 1 - \sqrt{1-a}$, then $0 < b < a$.

   d) Prove that if $3 < a < 5$ and $b = 2 + \sqrt{a-2}$, then $3 < b < a$.

7. The arithmetic mean of $a, b \in \mathbb{R}$ is $A(a,b) = (a+b)/2$, and the geometric mean of $a, b \in [0, \infty)$ is $G(a,b) = \sqrt{ab}$. If $0 \leq a \leq b$, prove that $a \leq G(a,b) \leq A(a,b) \leq b$. Prove that $G(a,b) = A(a,b)$ if and only if $a = b$.

8. Let $x \in \mathbb{R}$.

   a) Prove that $|x| \leq 2$ implies $|x^2 - 4| \leq 4|x-2|$.

   b) Prove that $|x| \leq 1$ implies $|x^2 + 2x - 3| \leq 4|x-1|$.

   c) Prove that $-3 \leq x \leq 2$ implies $|x^2 + x - 6| \leq 6|x-2|$.

   d) Prove that $-1 < x < 0$ implies $|x^3 - 2x + 1| < 1.25|x-1|$.

.....